

Unit 2 – Part I: Practice Sheet

Practice Sheet

1. Print numbers from 1 to n.

Sample Input/Output:

Input: 5

Output: 1 2 3 4 5

2. Print numbers from n to 1.

Sample Input/Output:

Input: 5

Output: 5 4 3 2 1

3. Print all even numbers from 1 to n.

Sample Input/Output:

Input: 10

Output: 2 4 6 8 10

4. Print all odd numbers from 1 to n.

Sample Input/Output:

Input: 9

Output: 1 3 5 7 9

5. Find the sum of numbers from 1 to n.

Sample Input/Output:

Input: 5

Output: 15

6. Find the sum of all odd numbers between 1 and n.

Sample Input/Output:

Input: 10

Output: 25

7. Print all numbers between 1 and n divisible by both 3 and 5.

Sample Input/Output:

Input: 50

Output: 15 30 45

8. Find the sum of all numbers between 1 and n divisible by 3.

Sample Input/Output:

Input: 10

Output: 18

9. Print the multiplication table of a number from 1 to 10.

Sample Input/Output:

Input: 5

Output: 5 10 15 20 25 30 35 40 45 50

10. Find the factorial of a number.

Sample Input/Output:

Input: 5

Output: 120

11. Print all factors of a given number.

Sample Input/Output:

Input: 12

Output: 1 2 3 4 6 12

12. Check whether a given number is prime.

Sample Input/Output:

Input: 17

Output: Prime Number

13. Print all prime numbers between 1 and n.

Sample Input/Output:

Input: 10

Output: 2 3 5 7

14. Find the sum of prime numbers between 1 and n.

Sample Input/Output:

Input: 10

Output: 17

15. Count the number of even digits in a number.

Sample Input/Output:

Input: 24853

Output: 3

16. Count the number of odd digits in a number.

Sample Input/Output:

Input: 24853

Output: 2

17. Find the first digit of a number.

Sample Input/Output:

Input: 58321

Output: 5

18. Find the last digit of a number.

Sample Input/Output:

Input: 58321

Output: 1

19. Check whether the digits of a number are in increasing order.

Sample Input/Output:

Input: 12349

Output: Increasing Order

20. Find the second largest digit of a number without sorting.

Sample Input/Output:

Input: 58321

Output: 5

21. Find the largest and second-largest digits in a number.

Sample Input/Output:

Input: 58321

Output: 8 5

22. Repeatedly calculate the sum of digits until a single-digit number is obtained.

Sample Input/Output:

Input: 9875

Output: 2

23. Find the frequency of every digit present in a number.

Sample Input/Output:

Input: 1223342

Output: 1:1, 2:3, 3:2, 4:1

24. Check whether a number is a strong number.

Sample Input/Output:

Input: 145

Output: Strong Number

25. Generate the Fibonacci series up to n terms.

Sample Input/Output:

Input: 6

Output: 0 1 1 2 3 5

26. Print the following pattern: *, **, ***, ****, *****.

Sample Input/Output:

Input: 5

Output: 5 rows, increasing stars

27. Print the following inverted pattern: ***** , **** , *** , ** , *.

Sample Input/Output:

Input: 5

Output: 5 rows, decreasing stars

28. Print the number pattern: 1, 12, 123, 1234, 12345.

Sample Input/Output:

Input: 5

Output: 5 rows

29. Print the number pattern: 1, 22, 333, 4444, 55555.

Sample Input/Output:

Input: 5

Output: 5 rows

30. Print a centered number palindrome pattern up to n rows.

Sample Input/Output:

Input: 5

Output: 5 rows